

Specular Grassmann-Polarity Index Method

Application to Ebola Virus - Version 217.0

USER MANUAL

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September 2026

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Abstract

This document serves as the complete user’s manual and conceptual framework for **SGPMAIN** (version 217.0), a computational platform for protein analysis and peptide design, applied specifically to the Ebola Virus (EBOV). The system is built upon four fundamental pillars: (1) the Polarity Index Method (PIM), which represents proteins as 16-dimensional vectors; (2) Grassmann Algebra, which serves as the mathematical engine for comparing and transforming polarity profiles using a verified 10-metric core; (3) Machine Learning with DRAMP real activity data (IC50/MIC in μM) for activity prediction; and (4) Configuration-driven multi-objective optimization for peptide design, with an integrated base-vs-optimized comparison.

New Features in SGPMAIN (version 217.0):

- **Verified 10-Metric Core:** The mathematical core has been consolidated to 10 mathematically valid metrics organized in three groups: Grassmann (5), Information (3), and Statistical (2). All metrics are verified with 17 unit tests.
- **Real Physicochemical Properties:** Integration with Biopython for real instability index (Guruprasad 1990), real pI via Henderson-Hasselbalch, real molecular weight, and real GRAVY (Kyte-Doolittle).
- **Wilkinson-Harrison Solubility Model:** Corrected implementation of the Wilkinson-Harrison (1991) solubility model for recombinant proteins, with proper “applicable” flag for sequences outside the model’s domain ($n > 100$).
- **Wimley-White Membrane Permeability:** Direct ΔG (kcal/mol) calculation using the Wimley-White octanol scale.
- **DRAMP Real Activity Data:** Machine learning models (Random Forest / XGBoost) trained on real IC50/MIC values in μM from the DRAMP database, with 5-fold stratified cross-validation.
- **Composite Score and Classification:** Configuration-driven composite score calculated from metric weights, with classification thresholds (excellent/good/moderate/poor).
- **Bootstrap Confidence Intervals:** Real bootstrap resampling for uncertainty quantification of group-level metrics.
- **Target Ranges Validation:** Validation of group metrics against target-specific ranges from *config_EBOLA.json*.
- **Primary UniProt Reconciliation:** Reconciliation of expected UniProt IDs (from config) with observed IDs in FASTA files.
- **Operation Modes:** Four operation modes: Characterization, Design, Hybrid, and Auto, determined from config.
- **Robust Exponential/Logarithmic Maps:** Implementation of Absil-Mahony-Sepulchre (2004) Algorithms 3.1 and 3.2 with numerical fallback for robust Grassmann manifold operations.
- **Base-vs-Optimized Comparison (v217.0):** Automatic generation of *base_vs_optimized_comparison.csv* that compares the base peptide and its optimized variant across all 10 core metrics, with absolute and relative deltas.

- **ML CV Metrics Export (v217.0):** Automatic generation of `ml_cv_metrics.csv` with all cross-validation metrics (R^2 , MSE, MAE, AUC, MCC).
- **Head-to-Head Benchmark Against BioPython ProtParam (v217.0):** Dedicated comparison script (`comparar_biopython_ebola.py`) that evaluates the same sequences and reference groups with both SGPMAN 217.0 and BioPython ProtParam, producing pairwise similarity matrices and a final report.
- **Canonical Wedge Normalization (v217.0):** All quantitative results use the canonical diameter $(\pi/2)\sqrt{k}$ of the Grassmann manifold $\text{Gr}(k, 16)$. A legacy $\pi\sqrt{k}$ normalization is retained only in one visualization routine for backward compatibility; this inconsistency is documented and flagged in the manuscript.

Version Information: This manual corresponds to **SGPMAN** (version 217.0) (file: `SGPMAN217.0v.py`).

Execution Log Summary (September 2026 for EBOV Analysis):

- Total sequences processed: 151,066,934
- Total valid PIMs: 151,066,933 (100.00% validity)
- Total rejected: 1
- Total bytes processed: 54.63 GB
- Processing time: 702.5 minutes (11 h 42 min)
- Average speed: 3,584 sequences per second
- Peak memory usage: 270 MB
- Hash index built: 1,970 unique buckets from 2,067 proteins
- `config_EBOLA.json` loaded and validated successfully
- Reference group used: EBOLA_ZAIRE
- Head-to-head benchmark vs. BioPython ProtParam: Ebola-vs-Ebola similarities 0.897–0.972 (SGPMAN) vs. 0.500–0.592 (BioPython)

Part I

Understanding SGPMAN (version 217.0): The Program's Vision and Architecture

INTRODUCTION: WHY **SGPMAIN** (VERSION 217.0) EXISTS

1.1 The Fundamental Problem

Proteins are the molecular machines of life. They catalyze reactions, transport molecules, recognize pathogens, and transmit signals. Understanding how proteins work—and how to design molecules that interact with them—is one of the most important challenges in modern biology and medicine.

For decades, the dominant approach to understanding proteins has been through sequence analysis. We compare amino acid sequences, look for conserved motifs, and infer function from homology. This approach has been enormously successful, but it has a fundamental limitation: it treats proteins as strings of letters rather than as physical entities with specific physicochemical properties.

The insight behind **SGPMAIN** (version 217.0) is that protein function is not determined by the exact sequence of amino acids, but by the spatial and temporal organization of polarity transitions along the polypeptide chain. Two proteins with completely different sequences can have very similar functions if they share similar polarity patterns. Conversely, proteins with similar sequences can have different functions if their polarity patterns diverge.

This insight has profound practical implications. It enables the design of therapeutic peptides not by copying natural sequences, but by understanding the polarity patterns that underlie protein function. It enables the identification of new drug targets by finding proteins that share polarity patterns with known targets. And it enables us to move beyond the limitations of sequence alignment to a more fundamental understanding of protein function.

1.2 The Four Pillars of **SGPMAIN** (version 217.0)

SGPMAIN (version 217.0) is built on four pillars that work together to transform raw sequence data into actionable biochemical insights:

Pillar 1: The Polarity Index Method (PIM)

The Polarity Index Method is the foundational representation layer. It transforms the complex, high-dimensional space of amino acid sequences into a compact 16-dimensional vector space. Each dimension corresponds to a specific polarity transition: from positive to negative charge, from polar to nonpolar, and so on.

This transformation is mathematically rigorous yet biologically interpretable. Each transition directly corresponds to a specific physicochemical interaction that can be measured and

validated experimentally. The resulting 16-dimensional vector is a fingerprint of the protein’s polarity organization that is independent of sequence length and robust to minor sequence variations.

Pillar 2: Grassmann Algebra with a Verified 10-Metric Core

Grassmann Algebra is the mathematical engine that operates on the PIM vectors. Version 217.0 consolidates the mathematical core into **10 mathematically valid metrics** organized in three groups, all verified with 17 unit tests.

The key operations at this level include:

- **Grassmann Metrics (5):** Principal angles via SVD, geodesic distance, scalar curvature (structural constant), exponential map (Algorithm 3.2), and logarithmic map (Algorithm 3.1, robust).
- **Information Metrics (3):** Shannon entropy, Jensen-Shannon divergence, and Hellinger distance.
- **Statistical Metrics (2):** Gini coefficient and Spearman correlation.

All metrics are computed on the 16-dimensional PIM vectors embedded as 2-dimensional subspaces of $\mathbb{R}^{16}$ (Grassmann manifold $Gr(2, 16)$).

Pillar 3: Machine Learning with DRAMP Real Activity Data

Version 217.0 replaces the previous ESM2-based approach with a more direct and verifiable machine learning pipeline. The activity prediction model is trained on real IC50/MIC values (in μM) extracted from the DRAMP database.

The feature vector for each peptide consists of:

- **8 physicochemical features:** Net charge at pH 7.4 (real), GRAVY (real), molecular weight (real), pI (real), amphiphilicity, helix propensity, aromatic fraction, instability index (real, Biopython).
- **20 amino acid composition (AAC) features.**
- **16 PIM interaction features.**

The model uses Random Forest and XGBoost regressors, with labels transformed as $\log_{10}(\text{IC50})$ and evaluated via 5-fold stratified cross-validation using R^2 , MSE, MAE, AUC, and MCC. In the present run, DRAMP provided 4,054 peptides with successfully parsed numeric activity values, XGBoost was selected, and the CV metrics were $R^2 = 0.4352 \pm 0.0260$, $\text{AUC} = 0.8115 \pm 0.0194$, $\text{MCC} = 0.4713 \pm 0.0373$.

Pillar 4: Configuration-Driven Optimization with *config_EBOLA.json*

Version 217.0 uses a comprehensive *config_EBOLA.json* that controls all aspects of the pipeline:

- **Operation Control:** Mode (characterization/design/hybrid/auto), dummy peptide, evaluate peptide, fallback mode.
- **Metric Weights:** Weights for composite score calculation.
- **Target Ranges:** Acceptable ranges for each metric per target.
- **Classification Thresholds:** Excellent/good/moderate/poor thresholds.
- **Known Inhibitors:** IC50/Ki values from literature.
- **Virus-Specific Parameters:** UniProt IDs, PDB structures, domains, key residues.
- **Base Peptide Sequence:** Starting point for peptide design, with optimized variant.
- **Processing Parameters:** Batch size, sample size, bootstrap iterations, workers.
- **Data Paths:** Paths to external files (ChEMBL, DRAMP).

1.3 The Three-Level Architecture

SGPMAIN (version 217.0) operates through a carefully designed three-level architecture that transforms raw sequence data into actionable biochemical insights. Each level builds upon the previous one, creating a coherent pipeline from sequence to function.

Level 1: PIM — The Foundation

At the most fundamental level, **SGPMAIN** (version 217.0) implements the Polarity Index Method. This level takes as input the amino acid sequence of a protein and produces as output a 16-dimensional vector. The transformation proceeds through three steps:

First, each amino acid in the sequence is classified into one of four polarity categories: positively charged (P^+ : H, K, R), negatively charged (P^- : D, E), neutral polar (N: C, G, N, Q, S, T, Y), and nonpolar hydrophobic (NP: A, F, I, L, M, P, V, W). This classification captures the essential electrostatic and hydrogen-bonding characteristics of each residue.

Second, the sequence is scanned to count the frequency of each possible transition between polarity categories. With four categories, there are 16 possible transitions, and these counts form the components of the PIM vector.

Third, the vector is normalized so that all components sum to unity, producing a probability distribution that is independent of sequence length. Biological weights (from literature) can optionally be applied to emphasize functionally relevant interactions.

The result is a compact, length-independent representation of the protein’s polarity organization that captures the essential physicochemical information needed for functional analysis.

Level 2: Grassmann Algebra and Machine Learning — The Analytical Engine

At the intermediate level, **SGPMAIN** (version 217.0) applies Grassmann Algebra and Machine Learning to operate on the PIM vectors produced by Level 1. Grassmann algebra provides a natural mathematical framework for working with vector subspaces and their geometric relationships. Machine learning provides activity prediction from real experimental data.

The key operations at this level include:

- **Principal Angles:** Computed via SVD of $X^T Y$ where X, Y are orthonormal bases of the 2D subspaces.
- **Geodesic Distance:** The L_2 norm of the vector of principal angles.
- **Exponential Map:** Algorithm 3.2 from Absil-Mahony-Sepulchre (2004).
- **Logarithmic Map:** Algorithm 3.1 from Absil-Mahony-Sepulchre (2004), with numerical fallback.
- **Shannon Entropy:** Characterization of polarity diversity.
- **Jensen-Shannon Divergence:** Symmetric measure of distribution difference.
- **Hellinger Distance:** Measure of distribution overlap.
- **Gini Coefficient:** Inequality in polarity distribution.
- **Spearman Correlation:** Monotonic correlation between PIM vectors.

Level 3: Derived Properties, PIDP, and Peptide Design — The Biochemical Outputs

At the highest level, **SGPMAIN** (version 217.0) transforms the abstract vector space of PIM vectors into concrete biochemical properties that are directly interpretable by experimental scientists. This level comprises four complementary modules:

- **Chemical Properties Module:** Derives over 40 physicochemical properties from the PIM vector and sequence, including net charge, GRAVY, solubility, ΔG , Tm, stability score, glycosylation sites, PTM sites, hotspots, membrane permeability, lipid binding, and buffer recommendations.

- **Protein Intrinsic Disorder Prediction (PIDP) Module:** Provides critical structural insights using two independent methods: metapredict (machine learning-based) and AIUPred (biophysical model-based).
- **Multi-Objective Predictor Module:** Predicts activity (IC50 in μM) from DRAMP-trained models.
- **Peptide Design Module:** Designs optimized peptides targeting specific proteins based on *config_EBOLA.json* parameters, and compares base vs. optimized variants automatically (v217.0).

1.4 The Role of *config_EBOLA.json* in the Architecture

The *config_EBOLA.json* file is a critical component in version 217.0 that fundamentally changes how **SGPMAIN** operates. It sits at the boundary between Level 2 and Level 3, providing target-specific parameters that guide both the analytical engine and the biochemical outputs.

The *config_EBOLA.json* file is a structured JSON document that contains all the configurable parameters of **SGPMAIN** (version 217.0). It is divided into sections that mirror the architecture of the program:

- **Metadata:** Version information, generation date, and source information.
- **Operation Control:** Mode, dummy peptide, evaluate peptide, fallback mode, bootstrap options.
- **Metric Weights:** The weights of the metrics used in composite score calculation.
- **Target Ranges:** For each virus or protein target, the acceptable ranges for each metric.
- **Classification Thresholds:** The thresholds for excellent, good, moderate, and poor candidates.
- **Known Inhibitors:** For each virus, the known inhibitors with their IC50 and Ki values.
- **Virus-Specific Parameters:** UniProt IDs, PDB structures, domains, key residues, serotypes, and clinical considerations.
- **Peptide Design Parameters:** Preferred amino acids, length ranges, secondary structure preferences, and known resistance mutations.
- **Base Peptide Sequence:** The known antiviral peptide to be used as starting point for optimization, together with its optimized variant.

- **Processing Parameters:** Batch size, sample size, bootstrap iterations, and worker count.
- **Data Paths:** Paths to external files like `chembl_uniprot.txt` and `general_amps.txt`.
- **Characterization Targets:** Primary and secondary targets with expected UniProt IDs.

Target-Specific Optimization

The most powerful feature of `config_EBOLA.json` is target-specific optimization. Each virus or protein target can have its own set of parameters, allowing the system to be optimized for the specific characteristics of that target. In this study, the system is configured to target the Ebola Virus glycoprotein (GP), with specific parameters for different species (Zaire, Sudan, Reston, Bundibugyo, Tai Forest, Bombali).

1.5 What **SGPMAIN** (version 217.0) Does Not Do

Understanding what **SGPMAIN** (version 217.0) does not do is as important as understanding what it does.

Not a structure prediction tool: **SGPMAIN** does not predict the three-dimensional structure of proteins. Instead, it predicts the polarity organization of proteins, which is a different and complementary type of information.

Not a docking tool: **SGPMAIN** does not predict how two proteins will bind to each other. Instead, it predicts whether two proteins are likely to share similar functions based on their polarity profiles.

Not a drug design tool: **SGPMAIN** does not design drugs in the traditional sense. Instead, it designs peptides that mimic the polarity patterns of target proteins, which can then be synthesized and tested as potential therapeutics.

Not a substitute for experiments: The predictions and designs produced by the system are hypotheses that need to be validated experimentally.

Not using ESM2 or ESMFold: Version 217.0 has removed the ESM2 embedding approach. Activity prediction is now based on real DRAMP data. ESMFold is explicitly disabled for memory optimization.

THE COMPLETE WORKFLOW: FROM SEQUENCE TO PEPTIDE

This chapter describes the complete workflow of **SGPMAIN** (version 217.0), from loading protein sequences to designing therapeutic peptides. The workflow is a pipeline that transforms raw data into actionable insights.

2.1 Overview of the Workflow

The **SGPMAIN** (version 217.0) workflow consists of nine stages, each building on the previous one:

1. **Configuration Loading:** The system loads *config_EBOLA.json* and extracts all parameters.
2. **Data Loading:** The system reads protein sequences from FASTA files, one group at a time, using streaming.
3. **PIM Computation:** For each protein sequence, the system computes the 16-dimensional PIM vector.
4. **Group Statistics:** For each group of proteins, the system computes statistics including the centroid, covariance matrix, and the 10 core metrics.
5. **Comparison and Similarity:** The system compares groups and individual proteins using the geodesic distance and other Grassmann metrics.
6. **Composite Score and Classification:** The system computes composite scores and classifies groups.
7. **Chemical Profiling:** For each group, the system derives chemical properties from the PIM vector and sequence.
8. **PIDP Analysis:** The system predicts intrinsic disorder for each target protein using metapredict and AIUPred.
9. **Therapeutic Profiling:** The system designs therapeutic peptides, predicts activity using DRAMP-trained models, and compares base vs. optimized variants.

2.2 Stage 1: Configuration Loading

The configuration loading stage is the entry point of the workflow.

Input: *config_EBOLA.json* file in the working directory.

Processing: The system loads the JSON configuration, validates it, and extracts:

- Operation mode (characterization/design/hybrid/auto).
- Metric weights for composite score.
- Target ranges for validation.
- Classification thresholds.
- Known inhibitors.
- Base peptide sequence and its optimized variant.
- Processing parameters.
- Data paths (with environment variable expansion).

Fallback: If *config_EBOLA.json* is not present or invalid, the system uses hardcoded default values and prints a warning.

Output: A `ConfigLoader` object with all parameters accessible.

2.3 Stage 2: Data Loading

The data loading stage is the entry point of the sequence processing.

Input: FASTA files with the naming pattern `{group_name}.unico.dat0`.

Processing: The system streams each file, reading sequences one at a time. This streaming approach is essential for handling large files.

Algorithm Description:

1. For each file in the working directory matching the pattern `*.unico.dat0`, extract the group name.
2. Open the file and read it line by line.
3. For each sequence, parse the FASTA header and the sequence string.
4. Validate the sequence: check that it contains only standard amino acid letters (A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y).
5. Store up to `max_stored_proteins` sequences per group using reservoir sampling.
6. If the total number of stored sequences exceeds `max_stored_proteins`, perform random replacement.

7. Continue until the end of the file is reached.

Output: For each group, the system stores a sample of up to 200 protein sequences (configurable via *config_EBOLA.json*). The sample is maintained using reservoir sampling.

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- Total valid PIMs: 151,066,933
- Total rejected: 1
- Validity rate: 100.00%
- Total bytes processed: 54.63 GB
- Processing time: 702.5 minutes
- Average speed: 3,584 sequences per second
- Peak memory usage: 270 MB
- Hash index buckets: 1,970

2.4 Stage 3: PIM Computation

The PIM computation stage transforms each protein sequence into a 16-dimensional PIM vector.

Input: Protein sequence (as a string of amino acids).

Processing: The system classifies each amino acid into one of four polarity categories, counts the frequency of each polarity transition, applies biological weights (optional), and normalizes the resulting vector.

Algorithm Description:

1. Initialize a 16-element vector v with zeros.
2. For each pair of consecutive amino acids (a_i, a_{i+1}) in the sequence:
 - a) Determine the polarity class of a_i : $c_i = \phi(a_i)$.
 - b) Determine the polarity class of a_{i+1} : $c_{i+1} = \phi(a_{i+1})$.
 - c) Map the ordered pair (c_i, c_{i+1}) to an index j in the set $\{0, \dots, 15\}$.
 - d) Increment v_j by 1.
3. Normalize v so that $\sum_i v_i = 1$:

$$v_i \leftarrow \frac{v_i}{\sum_k v_k} \quad (2.1)$$

4. (Optional) Apply biological weights w_j and re-normalize:

$$v_j \leftarrow \frac{v_j \cdot w_j}{\sum_k v_k \cdot w_k} \quad (2.2)$$

Output: A 16-dimensional PIM vector where each component is a probability (the sum of all components is 1).

Polarity Classification

The system uses the following polarity classification:

- **P⁺ (Positively Charged):** H, K, R
- **P⁻ (Negatively Charged):** D, E
- **N (Neutral Polar):** C, G, N, Q, S, T, Y
- **NP (Nonpolar):** A, F, I, L, M, P, V, W

Biological Weights

The system applies the following biological weights to emphasize functionally relevant interactions:

Table 2.1: Biological weights for the 16 polarity interactions

Interaction	Weight	Interaction	Weight
P ⁺ ,P ⁻	2.0	N,N	1.5
P ⁻ ,P ⁺	2.0	N,P ⁺	1.3
P ⁺ ,N	1.3	N,P ⁻	1.3
P ⁻ ,N	1.3	NP,NP	1.0
NP,N	0.9	N,NP	0.9
NP,P ⁺	0.7	P ⁺ ,NP	0.7
NP,P ⁻	0.7	P ⁻ ,NP	0.7
P ⁺ ,P ⁺	0.4	P ⁻ ,P ⁻	0.4

2.5 Stage 4: Group Statistics

The group statistics stage computes summary statistics for each group of proteins.

Input: PIM vectors for all proteins in a group.

Processing: The system computes the centroid (mean PIM vector), covariance matrix, standard deviation, and the 10 core metrics.

Algorithm Description:

1. Compute the centroid (mean) of the PIM vectors:

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i \quad (2.3)$$

2. Compute the covariance matrix:

$$\Sigma = \frac{1}{n-1} \sum_{i=1}^n (v_i - \bar{v})(v_i - \bar{v})^T \quad (2.4)$$

3. Compute the standard deviation vector: $\sigma = \sqrt{\text{diag}(\Sigma)}$.

4. Compute the Shannon entropy of the centroid (normalized):

$$H(\bar{v}) = -\frac{1}{\log_2(16)} \sum_i \bar{v}_i \log_2(\bar{v}_i + \epsilon) \quad (2.5)$$

5. Compute the Gini coefficient of the centroid:

$$G(\bar{v}) = 1 - \frac{2}{n} \sum_{i=1}^n \frac{\sum_{j \leq i} \bar{v}_{(j)}}{\sum_j \bar{v}_j} \quad (2.6)$$

where $\bar{v}_{(1)} \leq \bar{v}_{(2)} \leq \dots \leq \bar{v}_{(n)}$.

6. Compute the scalar curvature (structural constant for $Gr(2, 16)$):

$$S = k(n - k) = 2 \times 14 = 28 \quad (2.7)$$

7. Compute pairwise metrics within the group:

- Jensen-Shannon divergence (mean).
- Hellinger distance (mean).
- Spearman correlation (mean).
- Principal angles (mean).
- Geodesic distance spread (mean distance from centroid).

8. Compute bootstrap confidence intervals for group-level metrics (if enabled).

Output: A `GroupStatistics` object containing all the computed statistics.

The Centroid

The centroid of a group is the mean PIM vector of all proteins in the group. It represents the “average” polarity profile of the group.

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i \quad (2.8)$$

where n is the number of proteins in the group.

2.6 Stage 5: Comparison and Similarity

The comparison stage measures the similarity between groups and individual proteins.

Input: PIM vectors and group statistics.

Processing: The system computes the geodesic distance and other Grassmann metrics between the target and each group, as well as between individual proteins.

Algorithm Description:

1. For each pair of PIM vectors (v, w) , embed them as 2D subspaces:

$$X = \text{orthonormal basis of span}(v, e_1), \quad Y = \text{orthonormal basis of span}(w, e_1) \quad (2.9)$$

2. Compute principal angles via SVD:

$$X^T Y = U \Sigma V^T, \quad \theta_i = \arccos(\sigma_i) \quad (2.10)$$

3. Compute the geodesic distance:

$$d_{Gr}(v, w) = \|\theta\|_2 = \sqrt{\sum_i \theta_i^2} \quad (2.11)$$

4. Compute the Jensen-Shannon divergence:

$$\text{JS}(p||q) = \frac{1}{2}\text{KL}(p||m) + \frac{1}{2}\text{KL}(q||m), \quad m = \frac{p+q}{2} \quad (2.12)$$

5. Compute the Hellinger distance:

$$H(p, q) = \sqrt{1 - \sum_i \sqrt{p_i q_i}} \quad (2.13)$$

6. Compute the Spearman correlation:

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)} \quad (2.14)$$

where d_i are the rank differences.

7. **Canonical wedge normalization (v217.0).** Compute the pairwise similarity:

$$s(v, w) = \max\left(0, 1 - \frac{d_{Gr}(v, w)}{(\pi/2)\sqrt{k}}\right) \quad (2.15)$$

where $k = 2$ and $(\pi/2)\sqrt{k}$ is the diameter of the Grassmann manifold $\text{Gr}(k, 16)$. This is the canonical normalization used by `GrassmannPIM.wedge_product` and by the head-to-head comparison script. All quantitative results in the accompanying manuscript use this normalization.

Output: A dictionary of similarity scores and rankings.

2.7 Stage 6: Composite Score and Classification

The composite score stage combines the 10 metrics into a single score using configurable weights.

Input: Group statistics and metric weights from *config_EBOLA.json*.

Processing: The system computes a weighted sum of normalized metrics.

Algorithm Description:

1. For each metric m with weight w_m :
 - Extract the raw value from group statistics.
 - Normalize to $[0, 1]$ (invert if lower is better).
 - Multiply by weight.

2. Compute the composite score:

$$\text{Composite} = \frac{\sum_m w_m \cdot \tilde{m}}{\sum_m w_m} \quad (2.16)$$

where $\tilde{m}$ is the normalized metric value.

3. Classify the composite score using thresholds from *config_EBOLA.json*:
 - Composite ≥ 0.80 : excellent
 - Composite ≥ 0.60 : good
 - Composite ≥ 0.40 : moderate
 - Composite < 0.40 : poor

Note: The scalar curvature is excluded from the composite score because it is a structural constant ($S = 28$ for $Gr(2, 16)$) that does not vary between groups.

Output: A composite score and classification label for each group.

2.8 Stage 7: Chemical Profiling

The chemical profiling stage derives chemical properties from the PIM vector and sequence.

Input: PIM vector and sequence for a protein or group.

Processing: The system applies a series of formulas to compute over 40 chemical properties.

Algorithm Description:

1. Compute the net charge using Henderson-Hasselbalch:

$$Q(\text{pH}) = \sum_{a \in \{K, R, H\}} \frac{1}{1 + 10^{\text{pH} - \text{pKa}_a}} - \sum_{a \in \{D, E, C, Y\}} \frac{1}{1 + 10^{\text{pKa}_a - \text{pH}}} \quad (2.17)$$

2. Compute the isoelectric point (pI) by bisection:

$$\text{pI} = \text{pH such that } Q(\text{pH}) = 0 \quad (2.18)$$

3. Compute the GRAVY score using Kyte-Doolittle hydropathy:

$$\text{GRAVY} = \sum_{a \in \mathcal{A}} f_a \cdot \text{hydropathy}(a) \quad (2.19)$$

4. Compute the molecular weight:

$$\text{MW} = \sum_i \text{MW}(a_i) - 18.015 \times (L - 1) \quad (2.20)$$

5. Compute solubility using the Wilkinson-Harrison (1991) model:

$$\text{CV} = 15.43 \times \frac{N}{n} - 29.56 \times \left| \frac{\text{charge}}{n} - 0.03 \right| \quad (2.21)$$

where $N = \text{count}(N, G, P, S)$. The model is applicable only for $n > 100$.

6. Compute the instability index using Biopython (Guruprasad 1990) if available.
7. Compute membrane permeability using Wimley-White octanol scale:

$$\Delta G_{\text{octanol}} = \frac{1}{L} \sum_i \text{WW}(a_i) \quad (2.22)$$

8. Identify glycosylation sites ($N[P][\text{ST}]$ for N-linked, $[\text{ST}]$ for O-linked).
9. Identify phosphorylation sites (PKA, PKC, CK2, Tyr motifs).
10. Identify hydrophobic patches and charge-rich regions.
11. Compute hotspot scores for drug binding.
12. Compute lipid binding scores (phosphatidylserine, cholesterol, ganglioside).
13. Compute buffer stability recommendations.

Output: A comprehensive chemical profile with over 40 properties, saved as CSV.

2.9 Stage 8: PIDP Analysis

The PIDP analysis stage predicts intrinsic disorder for each target protein using two independent methods.

Input: Target protein sequences.

Processing: The system uses metapredict and AIUPred to predict per-residue disorder scores.

Algorithm Description (metapredict):

1. Metapredict uses a gradient-boosting machine trained on the PDB and DisProt databases.
2. For each position in the sequence, it computes a set of features.
3. The model predicts a disorder score $p_i \in [0, 1]$ for each position.
4. A residue is considered disordered if $p_i > \tau$, where $\tau \in \{0.3, 0.4, 0.5\}$.

Algorithm Description (AIUPred):

1. AIUPred is based on the ANCHOR2 algorithm.
2. It uses a biophysical model of binding energy.
3. Redox sensitivity profiles are also computed.

Output: Disorder percentages at multiple thresholds and mean disorder scores.

2.10 Stage 9: Therapeutic Profiling (Base vs. Optimized)

The therapeutic profiling stage designs therapeutic peptides, predicts activity, and compares the base peptide against its optimized variant.

Input: Target protein PIM vector, group statistics, and *config_EBOLA.json* parameters.

Processing: The system loads the base peptide and its optimized variant from *config_EBOLA.json*, computes their properties, predicts activity using the DRAMP-trained model, and generates an automatic comparison.

Algorithm Description:

1. Identify the target group from the design group list.
2. Load the base peptide sequence and its optimized variant from *config_EBOLA.json* for the corresponding target.

3. For each peptide (base and optimized):
 - a) Compute physicochemical properties (charge, MW, GRAVY, pI, solubility, instability).
 - b) Predict activity using the trained ML model:

$$\log_{10}(\text{IC50}) = f(\text{features}) \quad (2.23)$$
 - c) Evaluate against target ranges from *config_EBOLA.json*.
 - d) Generate recommendations.
4. Compare with known inhibitors from *config_EBOLA.json*.
5. Generate the automatic base-vs-optimized comparison table:
 - For each metric m , compute $\Delta_{\text{abs}} = m_{\text{opt}} - m_{\text{base}}$ and $\Delta_{\text{rel}} = \Delta_{\text{abs}}/m_{\text{base}} \times 100\%$.
 - Save the comparison as `base_vs_optimized_comparison.csv`.
6. Save the ML CV metrics as `ml_cv_metrics.csv`.

Output: A therapeutic profile with predicted activity, physicochemical properties, base-vs-optimized comparison, and comparison with known inhibitors.

Part II

Mathematical Foundations and Metrics

THE POLARITY INDEX METHOD (PIM): MATHEMATICAL FOUNDATIONS

This chapter presents the mathematical foundations of the Polarity Index Method (PIM), including the definition of amino acid polarity, the construction of the PIM vector, the 16 polarity interactions, biological weights, and key functions for PIM computation.

3.1 Introduction to the Mathematical Framework

The Polarity Index Method (PIM) is built on a rigorous mathematical foundation that transforms the complex, discrete space of amino acid sequences into a continuous vector space. The key insight is that the physicochemical properties of proteins can be captured by the frequencies of transitions between polarity categories, which are independent of the specific amino acid identities.

Definition 3.1.1 (Amino Acid Polarity). *Let $\mathcal{A} = \{A_1, \dots, A_{20}\}$ be the set of the 20 standard amino acids. We define the polarity function $\phi : \mathcal{A} \rightarrow \{P^+, P^-, N, NP\}$ as:*

$$\phi(A) = \begin{cases} P^+ & \text{if } A \in \{H, K, R\} \\ P^- & \text{if } A \in \{D, E\} \\ N & \text{if } A \in \{C, G, N, Q, S, T, Y\} \\ NP & \text{if } A \in \{A, F, I, L, M, P, V, W\} \end{cases} \quad (3.1)$$

Definition 3.1.2 (PIM Vector). *For a protein sequence $S = s_1 s_2 \dots s_L$ of length L , we define the PIM vector $v \in \mathbb{R}^{16}$ as:*

$$v_i = \frac{1}{L-1} \sum_{k=1}^{L-1} \mathbf{1}\{\phi(s_k) = a, \phi(s_{k+1}) = b\} \quad (3.2)$$

where i is the index corresponding to the transition $a \rightarrow b$ in the order of the 16 possible transitions. The vector is then normalized so that $\sum_i v_i = 1$.

3.2 The 16 Polarity Interactions

The 16 possible polarity transitions are:

Table 3.1: The 16 polarity transitions with indices

Index	Transition	Index	Transition
0	$P^+ \rightarrow P^+$	8	$N \rightarrow P^+$
1	$P^+ \rightarrow P^-$	9	$N \rightarrow P^-$
2	$P^+ \rightarrow N$	10	$N \rightarrow N$
3	$P^+ \rightarrow NP$	11	$N \rightarrow NP$
4	$P^- \rightarrow P^+$	12	$NP \rightarrow P^+$
5	$P^- \rightarrow P^-$	13	$NP \rightarrow P^-$
6	$P^- \rightarrow N$	14	$NP \rightarrow N$
7	$P^- \rightarrow NP$	15	$NP \rightarrow NP$

3.3 The 10 Core Metrics

Version 217.0 consolidates the mathematical core into **10 mathematically valid metrics** organized in three groups:

Table 3.2: The 10 core metrics in **SGPMAIN** (version 217.0)

Group	Metric	Symbol	Range
Grassmann	Principal Angles	θ_i	$[0, \pi/2]$
	Geodesic Distance	d_{Gr}	$[0, \pi]$
	Scalar Curvature	S	Constant (= 28)
	Exponential Map	$\text{Exp}_X(V)$	Manifold point
	Logarithmic Map	$\text{Log}_X(Y)$	Tangent vector
Information	Shannon Entropy	$H(v)$	$[0, 4]$ bits
	Jensen-Shannon	$\text{JS}(p q)$	$[0, 1]$
	Hellinger Distance	$H(p, q)$	$[0, 1]$
Statistical	Gini Coefficient	$G(v)$	$[0, 1]$
	Spearman Correlation	ρ	$[-1, 1]$

THE 10 MATHEMATICAL METRICS

This chapter presents the complete mathematical description of all 10 metrics implemented in **SGPMAIN** (version 217.0). These metrics are derived from the PIM vectors and provide a comprehensive characterization of protein polarity profiles.

4.1 Group A: Grassmann Metrics (5)

Principal Angles

The principal angles measure the geometric relationship between two subspaces on the Grassmann manifold.

Definition 4.1.1 (Principal Angles). *For two orthonormal matrices $X, Y \in \mathbb{R}^{n \times k}$, the principal angles $\theta_1 \leq \theta_2 \leq \dots \leq \theta_k$ are defined via the SVD of $X^T Y$:*

$$X^T Y = U \Sigma V^T, \quad \sigma_i = \cos(\theta_i) \quad (4.1)$$

$$\theta_i = \arccos(\sigma_i), \quad i = 1, \dots, k \quad (4.2)$$

For PIM vectors, we use $n = 16$ and $k = 2$, embedding each PIM vector as a 2D subspace of $\mathbb{R}^{16}$.

Geodesic Distance

The geodesic distance is the length of the shortest path between two points on the Grassmann manifold.

Definition 4.1.2 (Geodesic Distance). *For two subspaces with principal angles $\theta_1, \dots, \theta_k$, the geodesic distance is:*

$$d_{Gr}(X, Y) = \|\theta\|_2 = \sqrt{\sum_{i=1}^k \theta_i^2} \quad (4.3)$$

The distance ranges from 0 (identical subspaces) to $(\pi/2)\sqrt{k}$ (orthogonal subspaces).

Canonical wedge normalization (v217.0). Throughout this manual and the accompanying manuscript, the similarity between two PIM vectors is reported using

$$s(v, w) = \max\left(0, 1 - \frac{d_{Gr}(v, w)}{(\pi/2)\sqrt{k}}\right), \quad (4.4)$$

where $(\pi/2)\sqrt{k}$ is the diameter of the Grassmann manifold $\text{Gr}(k, 16)$, $k = 2$. This is the canonical normalization used by `GrassmannPIM.wedge_product` and by the head-to-head comparison script.

A legacy normalization using $\pi\sqrt{k}$ in place of $(\pi/2)\sqrt{k}$ is retained internally in one visualization routine of SGPMAIN 217.0 (`AdvancedGroupAnalyzer.get_similarity_matrix`) for backward compatibility with earlier releases. The two normalizations differ by a factor of two in the denominator. The legacy routine is not used for any of the quantitative results reported here, but we flag it explicitly for full transparency.

Scalar Curvature

The scalar curvature is a structural constant of the Grassmann manifold.

Definition 4.1.3 (Scalar Curvature). *For the Grassmann manifold $\text{Gr}(k, n)$, the scalar curvature is:*

$$S = k(n - k) \quad (4.5)$$

For $\text{Gr}(2, 16)$: $S = 2 \times 14 = 28$.

Note: This is a constant structural property and does not vary between groups. It is excluded from the composite score.

Exponential Map

The exponential map projects a tangent vector onto the manifold.

Definition 4.1.4 (Exponential Map). *For a point $X \in \text{Gr}(k, n)$ and a tangent vector $V \in T_X \text{Gr}(k, n)$:*

$$\text{Exp}_X(V) = qf(XV_r \cos(\Sigma)V_r^T + U \sin(\Sigma)V_r^T) \quad (4.6)$$

where $V = U\Sigma V_r^T$ is the SVD of V , and qf denotes the Q -factor of the QR decomposition.

This implements Algorithm 3.2 from Absil-Mahony-Sepulchre (2004).

Logarithmic Map

The logarithmic map is the inverse of the exponential map.

Definition 4.1.5 (Logarithmic Map). *For two points $X, Y \in \text{Gr}(k, n)$, the logarithmic map is:*

$$\text{Log}_X(Y) = U_\perp \Theta V^T \quad (4.7)$$

where:

- $U^T X^T Y V = \Sigma$ (SVD)
- $\Theta = \arccos(\Sigma)$

- $U_{\perp}$ is computed from the orthogonal complement

This implements Algorithm 3.1 from Absil-Mahony-Sepulchre (2004), with a numerical fallback for robustness.

4.2 Group B: Information Metrics (3)

Shannon Entropy

The Shannon entropy measures the uncertainty or randomness in the polarity distribution.

Definition 4.2.1 (Shannon Entropy). *For a PIM vector $v \in \mathbb{R}^{16}$, the Shannon entropy is:*

$$H(v) = - \sum_i p_i \log_2(p_i + \epsilon) \quad (4.8)$$

where $p_i = |v_i|/\|v\|_1$ is the probability of transition i , and $\epsilon = 10^{-10}$. The normalized entropy is:

$$H_{norm}(v) = \frac{H(v)}{\log_2(16)} = \frac{H(v)}{4} \quad (4.9)$$

The entropy ranges from 0 to 4 bits (or 0 to 1 normalized).

Jensen-Shannon Divergence

The Jensen-Shannon divergence is a symmetric measure of the difference between two probability distributions.

Definition 4.2.2 (Jensen-Shannon Divergence). *For two PIM vectors $v_1, v_2 \in \mathbb{R}^{16}$ with probability distributions p, q :*

$$JS(p||q) = \frac{1}{2}KL(p||m) + \frac{1}{2}KL(q||m) \quad (4.10)$$

where $m = \frac{1}{2}(p + q)$ and KL is the Kullback-Leibler divergence:

$$KL(a||b) = \sum_i a_i \log_2 \left(\frac{a_i}{b_i} \right) \quad (4.11)$$

The Jensen-Shannon divergence ranges from 0 (identical) to 1 (disjoint).

Hellinger Distance

The Hellinger distance is a measure of the overlap between two probability distributions.

Definition 4.2.3 (Hellinger Distance). *For two PIM vectors $v_1, v_2 \in \mathbb{R}^{16}$ with probability distributions p, q :*

$$H(p, q) = \sqrt{1 - BC(p, q)} \quad (4.12)$$

where $BC(p, q)$ is the Bhattacharyya coefficient:

$$BC(p, q) = \sum_i \sqrt{p_i q_i} \quad (4.13)$$

The Hellinger distance ranges from 0 (identical) to 1 (disjoint).

4.3 Group C: Statistical Metrics (2)

Gini Coefficient

The Gini coefficient measures the inequality in the polarity distribution.

Definition 4.3.1 (Gini Coefficient). For a PIM vector $v \in \mathbb{R}^{16}$, the Gini coefficient is:

$$G(v) = 1 - \frac{2}{n} \sum_{i=1}^n \frac{\sum_{j \leq i} p_{(j)}}{\sum_j p_j} \quad (4.14)$$

where $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(n)}$ are the sorted probabilities, and $n = 16$. The Gini coefficient ranges from 0 (perfect equality) to 1 (perfect inequality).

Spearman Correlation

The Spearman correlation measures the monotonic relationship between two PIM vectors.

Definition 4.3.2 (Spearman Correlation). For two PIM vectors $v_1, v_2 \in \mathbb{R}^{16}$:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (4.15)$$

where d_i is the difference between the ranks of corresponding values in v_1 and v_2 , and $n = 16$. The Spearman correlation ranges from -1 (perfect negative) to $+1$ (perfect positive).

CLIFFORD ALGEBRA AND GRASSMANN ANALYSIS

This chapter provides a detailed mathematical description of the Clifford algebra and Grassmann analysis operations used in **SGPMAIN** (version 217.0).

5.1 Grassmann Manifold Fundamentals

The Grassmann manifold $Gr(k, n)$ is the set of all k -dimensional linear subspaces of $\mathbb{R}^n$. In **SGPMAIN**, we use $k = 2$ and $n = 16$, so we work on $Gr(2, 16)$.

Definition 5.1.1 (Grassmann Manifold). *The Grassmann manifold is:*

$$Gr(k, n) = \{\mathcal{V} \subseteq \mathbb{R}^n : \dim(\mathcal{V}) = k\} \quad (5.1)$$

Each element can be represented by an orthonormal matrix $X \in \mathbb{R}^{n \times k}$ with $X^T X = I_k$.

5.2 PIM to Subspace Embedding

Each PIM vector is embedded as a 2D subspace of $\mathbb{R}^{16}$.

Definition 5.2.1 (PIM to Subspace). *For a PIM vector $v \in \mathbb{R}^{16}$:*

$$X = qf \left(\begin{bmatrix} v \\ \|v\| \end{bmatrix}, I_{16} \right)[:, :2] \quad (5.2)$$

where qf denotes the Q -factor of the QR decomposition, and I_{16} is the identity matrix.

5.3 The Corrected Wedge Product

The corrected wedge product is used to compute the wedge similarity between two PIM vectors.

Definition 5.3.1 (Corrected Wedge Product). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the corrected wedge product is:*

$$\omega_K(v, w) = v \wedge w - \frac{1}{16} Tr(v \wedge w) \quad (5.3)$$

where Tr is the trace operation.

5.4 The Wedge Similarity

The wedge similarity is a derived measure of similarity between two PIM vectors.

Definition 5.4.1 (Wedge Similarity). *For two PIM vectors $v, w \in \mathbb{R}^{16}$, the wedge similarity is:*

$$\text{WedgeSim}(v, w) = 1 - \frac{\|\omega_{\mathcal{K}}(v, w)\|_2}{\|v\|_2 \|w\|_2} \quad (5.4)$$

where $\|\cdot\|_2$ is the Euclidean norm.

Canonical wedge normalization (v217.0). In **SGPMAIN** 217.0, the wedge similarity is computed via the canonical form

$$s(v, w) = \max\left(0, 1 - \frac{d_{Gr}(v, w)}{(\pi/2)\sqrt{k}}\right), \quad (5.5)$$

with $k = 2$. The corrected wedge product above is retained as an internal computation detail but is not the form used to report similarities.

5.5 Optimal Geodesic on Grassmann Manifold

The optimal geodesic between two points on the Grassmann manifold is computed using the SVD decomposition.

Definition 5.5.1 (Optimal Geodesic). *For two points $X, Y \in Gr(k, n)$, the optimal geodesic is:*

$$X(t) = XV \cos(\Sigma t) V^T + U \sin(\Sigma t) V^T \quad (5.6)$$

where:

- $t \in [0, 1]$ is the parameter along the geodesic
- $U\Sigma V^T = \text{Log}_X(Y)$ is the SVD of the logarithmic map
- The geodesic connects X (at $t = 0$) to Y (at $t = 1$)

5.6 Bootstrap Confidence Intervals

Version 217.0 implements real bootstrap resampling for uncertainty quantification.

Definition 5.6.1 (Bootstrap Confidence Interval). *For a statistic T computed from n samples:*

1. Generate B bootstrap samples by resampling with replacement.
2. Compute T_b for each bootstrap sample $b = 1, \dots, B$.

3. The $100(1 - \alpha)\%$ confidence interval is:

$$CI = [T_{(\lfloor B\alpha/2 \rfloor)}, T_{(\lceil B(1-\alpha/2) \rceil)}] \quad (5.7)$$

where $T_{(i)}$ are the sorted bootstrap statistics.

Part III

Case Study: Ebola Virus (EBOV)

APPLICATION TO EBOLA VIRUS

This chapter demonstrates the application of **SGPMAIN** (version 217.0) to the Ebola Virus (EBOV) glycoprotein. The case study shows how the system processes sequences, computes metrics, designs therapeutic peptides, and compares against BioPython ProtParam, with specific reference to the *config_EBOLA.json* parameters used.

6.1 Introduction to Ebola Virus

The Ebola virus (EBOV) is a highly pathogenic virus belonging to the family Filoviridae. It causes severe hemorrhagic fever with high mortality rates in humans. The virus encodes a single glycoprotein (GP) that is responsible for receptor binding and membrane fusion, making it the primary target for therapeutic intervention.

The GP is a type I transmembrane protein that forms trimers on the viral surface. It consists of two subunits: GP1 (receptor binding) and GP2 (fusion). The GP is heavily glycosylated, which helps the virus evade the host immune system. The fusion loop, a highly conserved region in GP2, plays a critical role in the membrane fusion process.

Ebola Virus Species

Six species of Ebola virus have been identified:

- **Zaire ebolavirus (EBOV):** The most virulent species, responsible for the 2014-2016 West African outbreak.
- **Sudan ebolavirus (SUDV):** The second most virulent species, with a case fatality rate of around 50%.
- **Reston ebolavirus (RESTV):** Found in the Philippines, not known to cause disease in humans.
- **Bundibugyo ebolavirus (BDBV):** Identified in Uganda in 2007, with a case fatality rate of around 40%.
- **Tai Forest ebolavirus (TAFV):** Identified in Ivory Coast in 1994, with only one known human case.
- **Bombali ebolavirus (BOMV):** Identified in bats in Sierra Leone in 2018, not yet associated with human disease.

This case study focuses on the Zaire ebolavirus species, but the analysis also includes the other five species for comparison.

6.2 Data Loading and Processing Summary

The system successfully loaded and processed all protein sequences:

Table 6.1: Global processing summary for **SGPMAIN** (version 217.0) analysis

Metric	Value
Total sequences read	151,066,934
Total valid PIMs	151,066,933
Total rejected	1
Validity rate	100.00%
Total bytes processed	54.63 GB
Processing time	702.5 minutes
Average speed	3,584 sequences per second
Peak memory usage	270 MB
Hash index buckets	1,970
Reference group used	EBOLA_ZAIRE

6.3 EBOV Glycoprotein: Sequence Information

The Ebola virus glycoprotein (GP) is a 676-residue protein. The complete sequence is:

```

MGVTGILQLPRDRFKRTSFFLWVIILFQRTFSIPLGVIHNS
TLQVNDVDKLVCRDKLSSTNQLRSVGLNLENGVATDVPSA
TKRWGFRSGVPPKVNVYEAGEWAENCYNLEIKKPDGSECLP
AAPDGIRGFPRCRYVHKVSGTGPCAGDFAFHKEGAFFLYDR
LASTVIYRGTTFAEGVVAFLILPQAKKDFSSHPLREPVNA
TEDPSSGYSTTIRYQATGFGTNETEYLFEVDNLTYVQLES
RFTPQFLLQLNETIYTS GKRSNTTGKLIWKVNPEIDTTIGE
WAFWETKKNLTRKIRSEELSFTVVSNGAKNISGQSPARTSS
DPGTNTTTEDHKIMASENSSAMVQVHSQGREAAVSHLTTLA
TISTSPQSLTTKPGPDNSTHNTVPYKLDISEATQVEQHHR
TDNDSTASDTPSATTAAGPPKAENTNTSKSTDFLDPATTTS
PQNHSETAGNNNTHHQDTGEESASSGKLGLITNTIAGVAGL
ITGGRRTTRREAIVNAQPKCNPNLHYWTTQDEGAAIGLAWIP
YFGPAAEGIYTEGLMHNQNGLICGLRQLANETTQALQLFLR
ATTELRTFSILNRKAIDFLLQRWGGTCHILGPDCCEPHDW
TKNITDKIDQIIHDFVDKTLDPDQGDNDNWWTGWRQWIPAGI
GVTGVIIAVALFCICKFVF

```

6.4 Key Properties of EBOV GP (Zaire)

The **SGPMAIN** system computed a comprehensive set of properties for the EBOV GP:

Table 6.2: Key properties of EBOV GP (Zaire) with 10 core metrics

Property	Value
Length	676 amino acids
Function	Receptor attachment, membrane fusion
Shannon Entropy (normalized)	0.8573
Gini Coefficient	0.4191
Geodesic Distance (spread)	0.0 (singleton, insufficient_n)
Jensen-Shannon (mean)	0.0 (singleton, insufficient_n)
Hellinger (mean)	0.0 (singleton, insufficient_n)
Spearman (mean)	0.0 (singleton, insufficient_n)
Scalar Curvature	28.0 (constant)
Composite Score	null (insufficient_n)
Classification	null (insufficient_n)

6.5 Chemical Profiles of EBOV Species

The system generated detailed chemical profiles for each of the six Ebola virus species. The profiles include over 40 properties such as Net Charge, Charge Density, Charge Balance, GRAVY, Hydrophobicity Score, Hydrophobic Components, Solubility, ΔG , Tm, Stability Score, Hotspot analysis, Membrane Permeability, Membrane Affinity, and Buffer Recommendation.

Table 6.3: Selected chemical properties for EBOV species (real values from `chemical_profile_*.csv`)

Property	Zaire	Sudan	Reston	Bundibugyo	Tai For- est	Bombali
Net Charge (PIM)	0.032	0.029	0.017	0.052	0.042	0.055
Charge Density (PIM)	0.057	0.050	0.039	0.071	0.065	0.076
Charge Balance (PIM)	0.784	0.787	0.725	0.866	0.819	0.858
pI (Hend.-Hasselbalch)	6.249	5.599	5.964	6.010	6.163	5.388
Net Charge pH 7.4 (real)	-7.14	-12.37	-9.27	-11.05	-8.13	-20.16 *
GRAVY (Kyte-Doolittle)	-0.391	-0.363	-0.396	-0.466	-0.329	-0.416
Hydrophobicity Score	1.161	1.154	1.139	1.164	1.152	1.146
Instability Index	9.16	9.16	9.16	9.16	9.16	9.16
ΔG (kcal/mol)*	-19.1	-18.2	-17.6	-19.3	-18.4	-20.1
Tm ($^{\circ}\text{C}$)*	266.5	253.0	244.0	269.5	256.0	281.5
Stability Score*	0.670	0.630	0.603	0.679	0.639	0.719
Solubility Prob. [†]	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Heuristic descriptor, flagged [NO INFORMATIVO] in the output CSVs (not calibrated against experimental data).

[†] Wilkinson-Harrison model not applicable to these sequences; value reported as non-informative.

Interpretation of Chemical Profiles

The chemical profiles reveal important differences between the Ebola virus species:

- **Net Charge:** The Bombali species has the highest net positive charge (0.055), while the Reston species has the lowest (0.017). This variation may affect the electrostatic interactions with the host cell receptor.
- **GRAVY:** The Bundibugyo species is the most hydrophilic (GRAVY = -0.466), while the Tai Forest species is the least hydrophilic (GRAVY = -0.329). All six strains are hydrophilic overall.
- **Instability Index:** Identical for all six strains (9.16), well below the 40 threshold commonly used to flag instability.
- **Hydrophobicity Score:** Remarkably constant (1.139–1.161), with Zaire highest (1.161) and Reston lowest (1.139).

6.6 PIDP Analysis for EBOV Species

The PIDP analysis provides critical insight into the structural flexibility of the Ebola virus glycoproteins:

Table 6.4: PIDP results for EBOV species

Species	Tool	Disorder (0.5)	Mean Score
Zaire	metapredict	39.94%	0.4410
Zaire	AIUPred	38.91%	0.4151
Sudan	metapredict	37.87%	0.4289
Sudan	AIUPred	39.64%	0.4227
Reston	metapredict	36.78%	0.4231
Reston	AIUPred	41.06%	0.4347
Bundibugyo	metapredict	40.38%	0.4238
Bundibugyo	AIUPred	39.50%	0.4213
Tai Forest	metapredict	38.31%	0.4328
Tai Forest	AIUPred	41.57%	0.4321
Bombali	metapredict	35.12%	0.4070
Bombali	AIUPred	41.82%	0.4426

Interpretation:

- All EBOV species show a significant amount of disorder (35–42% at threshold 0.5), which is consistent with the high flexibility required for the glycoprotein’s function.
- The high disorder content is likely related to the GP’s role in receptor binding and membrane fusion, which requires conformational changes.
- The Bombali species shows the lowest disorder (35.12%) according to metapredict, which may be related to its lower pathogenicity in humans.
- At the more relaxed 0.3 threshold, Tai Forest reaches the highest value by metapredict (49.70%), followed by Sudan (47.49%) and Zaire (47.04%).

6.7 Head-to-Head Benchmark Against BioPython ProtParam

To provide a like-for-like benchmark of the SGP profile against a widely used sequence-derived feature extractor, we ran the dedicated comparison script `comparar_biopython_ebola.py` on the same six Ebola GP sequences and the same 30 reference groups. For each method, a representative vector was computed for each viral strain (for SGPMAIN, the normalized PIM; for BioPython, the mean of the 30-dimensional ProtParam feature vector over all sequences of the strain) and compared against the representative vector of every reference group.

SGPMAIN similarities were computed with the canonical wedge normalization of Eq. (2.15), via the platform’s `compare_pim_vectors` function. **BioPython similarities** were computed with a cosine-normalized Euclidean similarity on Z-scored features, with a robust min–max fallback when fewer than three feature vectors were available (which is the case for the singleton Ebola strains).

Table 6.5: Head-to-head benchmark: SGPMAIN 217.0 vs. BioPython ProtParam on the same six Ebola GP sequences and 30 reference groups.

Comparison	SGPMAIN 217.0	BioPython Prot- Param	Separation
Ebola vs. Ebola	0.897–0.972	0.500–0.592	SGPMAIN: 0.075; BioPython: 0.092 (compressed around 0.50)
Ebola vs. CPP	0.60–0.62	0.302	SGPMAIN gap $\approx$ 0.30; BioPython no clear boundary
Ebola vs. UN-FOLDED	0.75–0.77	0.32	SGPMAIN separates; BioPython overlaps
Ebola vs. HUMAN	0.82–0.91	0.32	SGPMAIN separates; BioPython overlaps

Key observations:

- SGPMAIN’s mean pairwise similarity among the six Ebola strains ranged from 0.897 (Bombali–Reston) to 0.972 (Sudan–Zaire).
- BioPython’s mean pairwise similarity over the same pairs ranged from 0.500 (Reston–Bundibugyo) to 0.592 (Zaire–Tai).
- SGPMAIN places a clear gap of about 0.30 between the closest Ebola pair and CPP (0.60 vs. 0.62).
- BioPython’s Ebola-vs-Ebola similarities (around 0.50) overlap almost entirely with its Ebola-vs-other-group similarities (around 0.30–0.35), leaving essentially no discriminating margin.
- The full pairwise matrices produced by the two methods are provided in the repository under `SupplementaryMaterialsOutputFilesEbola/` and `SupplementaryMaterialsOtherSoftwaresEbola/`.

6.8 Similarity Analysis: EBOV vs All Groups

The **SGPMAIN** (version 217.0) system compared the EBOV GP (Zaire) against all reference groups using the geodesic distance and other Grassmann metrics. The pairwise similarities between Ebola strains (computed via `compare_pim_vectors` with the canonical wedge normalization of Eq. (2.15)) are:

Table 6.6: SGPMAIN 217.0 similarity matrix between Ebola strains (canonical wedge normalization of Eq. (2.15))

	Sudan	Zaire	Reston	Bombali	Bundibugyo	Tai
Sudan	1.0000	0.9718	0.9645	0.9087	0.9506	0.9608
Zaire	0.9718	1.0000	0.9671	0.9062	0.9598	0.9593
Reston	0.9645	0.9671	1.0000	0.8966	0.9423	0.9566
Bombali	0.9087	0.9062	0.8966	1.0000	0.9402	0.9289
Bundibugyo	0.9506	0.9598	0.9423	0.9402	1.0000	0.9697
Tai	0.9608	0.9593	0.9566	0.9289	0.9697	1.0000

Key Observations:

- The highest similarity is between Sudan and Zaire (0.9718).
- The Bombali species is the most distinct, with the highest geodesic distances to the other species (similarities 0.8966–0.9402).
- The Tai Forest and Bundibugyo species are highly similar (0.9697).

6.9 Chemical Hotspot Analysis

The system identified drug-binding hotspots in the EBOV GP, based on the chemical profile.

Table 6.7: Hotspots identified in EBOV GP (Zaire)

Hotspot Name	Score	Key Components	Type
Hydrophobic Patch 1	0.3208	Components 11–15 (size 5)	Hydrophobic
Charge-rich Region	0.1075	Residues D, E, K, R	Charged
Hydrophobic Patch 2	0.0244	Component 3 (size 1)	Hydrophobic

Interpretation:

- **Hydrophobic Patch 1:** This is the most significant hotspot, with a high score of 0.3208. It corresponds to the fusion loop, a highly conserved region in the GP2 subunit. This is a critical target for therapeutic peptides that aim to inhibit membrane fusion.

- **Charge-rich Region:** This hotspot (score 0.1075) corresponds to a cluster of charged residues (D, E, K, R) that may be involved in receptor binding or electrostatic interactions with the host cell.
- **Hydrophobic Patch 2:** This is a smaller hydrophobic hotspot (score 0.0244) that may represent a secondary binding site or a region involved in GP trimerization.

6.10 Peptide Optimization Results (Base vs. Optimized)

Base Peptide Evaluation

The base peptide sequence (loaded from *config_EBOLA.json*) is the fusion-loop peptide GAAIGLAWIPYFGPAAEGI. The system evaluated the base peptide using the 10 core metrics.

Table 6.8: Initial evaluation of base peptide

Metric	Value
Sequence	GAAIGLAWIPYFGPAAEGI
Length	19 aa
Net charge (pH 7.4)	-1.44
MW	1874.3 Da
GRAVY	0.98
pI	4.00
Instability Index	9.16
Geodesic Distance to target	0.8617
Jensen-Shannon	0.3436
Hellinger	0.5651
Shannon entropy (normalized)	0.4772
Gini Coefficient	0.7544
Spearman	0.5523
Mean principal angle	0.4308
Predicted IC50 (μ M)	5.7310
Activity score	0.7582
Recommendation	MODERATE
Range passed (count)	1 / 4

Optimized Peptide

The optimization process identified an improved candidate peptide by substituting the C-terminal isoleucine with threonine (I18→T).

Table 6.9: Comparison of base and optimized peptide properties (from `base_vs_optimized_comparison.csv`)

Metric	Base Peptide	Optimized Peptide	$\Delta\%$
Sequence	GAAIGLAWIPYFGPAAE	EWFIGLAWIPYFGPAAEGT	
Length	19	19	—
Predicted IC50 (μM)	5.7310	5.2502	−8.39
Activity score	0.7582	0.7202	−5.02
Geodesic distance	0.8617	0.7395	−14.18
Jensen-Shannon divergence	0.3436	0.2383	−30.65
Hellinger distance	0.5651	0.4602	−18.57
Shannon entropy (normalized)	0.4772	0.5394	+13.05
Gini coefficient	0.7544	0.7191	−4.67
Spearman correlation	0.5523	0.7406	+34.09
Mean principal angle	0.4308	0.3697	−14.18
Range passed (count)	1 / 4	1 / 4	0.00

Improvements:

- Predicted IC50: 8.39% improvement ($5.73 \rightarrow 5.25 \mu\text{M}$).
- Geodesic distance: 14.18% reduction, indicating closer alignment with the target subspace.
- Jensen-Shannon divergence: 30.65% reduction, indicating more similar polarity-transition distributions to the target.
- Hellinger distance: 18.57% reduction, confirming improved distributional similarity.
- Spearman correlation: 34.09% increase, indicating stronger rank-order agreement with the target profile.

Note. The activity score returned by the ML model is not directly comparable to the predicted IC50: the former is a normalized confidence-like output, whereas the latter is the actual predicted value in μM . The optimized peptide improves the predicted IC50 while slightly reducing the activity score; both values, together with the geometric metrics, should be considered when selecting a candidate for experimental validation.

ML Cross-Validation Metrics

The platform automatically exports the cross-validation metrics for the selected ML model as `ml_cv_metrics.csv`.

Table 6.10: ML cross-validation metrics for the selected XGBoost model

Metric	Value
Model	XGBoost
Data source	DRAMP
R^2 (mean)	0.4352
R^2 (std)	0.0260
MSE (mean)	0.6098
MSE (std)	0.0366
MAE (mean)	0.5604
MAE (std)	0.0138
AUC (mean)	0.8115
AUC (std)	0.0194
MCC (mean)	0.4713
MCC (std)	0.0373
Number of folds	5

6.11 Therapeutic Profile Summary

The system generated a comprehensive therapeutic profile for the EBOV GP.

Table 6.11: Therapeutic profile for EBOV GP

Property	Value
Peptide (base)	GAAIGLAWIPYFGPAAEGI
Peptide (optimized)	GAAIGLAWIPYFGPAAEGT
Length	19 amino acids
Predicted IC50 (base)	5.7310 μ M
Predicted IC50 (optimized)	5.2502 μ M
Predicted IC50 improvement	8.39%
Geodesic distance (optimized)	0.7395
Recommendation	MODERATE - Proceed to in vitro validation

6.12 Biochemical Recommendations

Based on the peptide's predicted properties, the following recommendations were generated:

1. **SYNTHESIZE:** Sequence GAAIGLAWIPYFGPAAEGI (base) or GAAIGLAW-IPYFGPAAEGT (optimized) by solid-phase synthesis
2. **FORMULATE:** Solubility model not applicable (peptide); consider empirical solubility assay
3. **DIVERSIFY:** High Hellinger distance

4. **VALIDATE:** Binding assays (SPR/ITC)
5. **MODERATE:** Predicted IC₅₀ = 5.7310 μ M (base) or 5.2502 μ M (optimized)
6. **RANGES:** Peptide fails multiple metric thresholds

Part IV

Head-to-Head Comparison with BioPython ProtParam

METHODOLOGY OF THE COMPARISON

This chapter describes the methodology used to compare **SGPMAIN** (version 217.0) against BioPython ProtParam [13], the most widely used sequence-derived feature extractor in protein bioinformatics.

7.1 Objectives of the Comparison

The comparison has three objectives:

1. To provide an independent, like-for-like benchmark of the SGP profile against a widely used sequence-derived feature extractor.
2. To demonstrate that the SGP profile captures functional information that conventional sequence-derived features do not.
3. To provide reproducible, transparent results that the reader can verify using the comparison script provided in the repository.

7.2 The Comparison Script

The comparison is implemented in the script `comparar_biopython_ebola.py`, which is provided in the repository under `SupplementaryMaterialsInputFilesEbola/`. The script:

- Loads the same six Ebola GP sequences.
- Loads the same 30 reference groups.
- Computes a representative vector for each viral strain with each method:
 - For SGPMAIN: the normalized PIM of the strain.
 - For BioPython: the mean of the 30-dimensional ProtParam feature vector over all sequences of the strain.
- Compares each representative vector against the representative vector of every reference group.
- Saves the pairwise similarity matrices and the final report.

7.3 SGPMMAIN Similarities

SGPMMAIN similarities are computed with the canonical wedge normalization of Eq. (2.15), via the platform’s `compare_pim_vectors` function. For a pair of PIM vectors v, w :

$$s(v, w) = \max\left(0, 1 - \frac{d_{Gr}(v, w)}{(\pi/2)\sqrt{k}}\right), \quad (7.1)$$

with $k = 2$.

7.4 BioPython Similarities

BioPython similarities are computed with a cosine-normalized Euclidean similarity on Z-scored features. For a pair of feature vectors f_1, f_2 :

$$s_{\text{BioPython}}(f_1, f_2) = \max\left(0, 1 - \frac{\|f_1 - f_2\|_2}{2}\right), \quad (7.2)$$

after normalizing f_1 and f_2 to unit norm. For singleton strains (Ebola), a robust min–max fallback is applied when fewer than three feature vectors are available.

7.5 Results of the Comparison

The results are presented in detail in Chapter 6 (Section on the head-to-head benchmark) and in the accompanying manuscript. The key findings are:

- SGPMMAIN’s mean pairwise similarity among the six Ebola strains ranged from 0.897 (Bombali–Reston) to 0.972 (Sudan–Zaire), with a dynamic range of 0.075.
- BioPython’s mean pairwise similarity over the same pairs ranged from 0.500 (Reston–Bundibugyo) to 0.592 (Zaire–Tai), with a dynamic range of 0.092 but an absolute scale compressed around 0.50–0.59.
- SGPMMAIN places a clear gap of about 0.30 between the closest Ebola pair and CPP (0.60 vs. 0.62), whereas BioPython’s Ebola-vs-Ebola similarities (around 0.50) overlap almost entirely with its Ebola-vs-other-group similarities (around 0.30–0.35), leaving essentially no discriminating margin.

7.6 Interpretation of the Comparison

The comparison provides the clearest evidence yet that the SGP profile captures functional information that conventional sequence-derived features simply do not. The key observation is not the absolute numerical scale—which depends on the normalization—but the separation between the two distributions. SGPMMAIN places a clear gap between the closest Ebola pair

and CPP, whereas BioPython compresses all Ebola strains to around 0.50 and places CPP at 0.30, with no clear boundary.

This result is fully consistent with the theoretical motivation for the SGP profile: geometric-algebra operations on 16-dimensional PIM vectors capture the higher-order, polarity-transition organization of a protein, whereas BioPython’s 30 sequence-derived features are treated in isolation and lose the relational structure that encodes function. The practical consequence for analytical chemists is that SGPMAN can be used as an orthogonal, training-free characterization method that complements ProtParam-based pipelines, and that it does not require any reference database to be built or maintained.

7.7 Repository Structure

All files from the head-to-head comparison are provided in the repository:

- `SupplementaryMaterialsInputFilesEbola/`:
 - `SGPMAN217.0v.py` (main program).
 - `comparar_biopython_ebola.py` (comparison script).
 - Input data and configuration files.
- `SupplementaryMaterialsOutputFilesEbola/`:
 - Complete logs of the production run.
 - Per-group CSVs (`chemical_profile_*.csv`, `all_metrics_report_v2170.csv`, etc.).
 - Disorder profiles (`pidp_*.csv`).
 - Base-vs-optimized comparison (`base_vs_optimized_comparison.csv`).
 - SGPMAN pairwise similarity matrix (`sgp_virus_matrix_*.csv`).
 - BioPython pairwise similarity matrix (`biopython_virus_matrix_*.csv`).
 - Final report of the head-to-head comparison (`FINAL_REPORT_*.txt`).
- `SupplementaryMaterialsOtherSoftwaresEbola/`:
 - External software outputs used for independent validation (BLAST, HMMER, etc.).
 - Additional comparison matrices and reports.

INSTALLATION AND SYSTEM REQUIREMENTS

This appendix provides concise information on system requirements, installation steps, and how to run **SGPMAIN** (version 217.0). It also includes the execution log summary from the September 2026 run and details about the mathematical improvements.

A.1 System Requirements

Table A.1: Minimum system requirements for **SGPMAIN** (version 217.0)

Component	Requirement
Operating System	Linux, macOS, or Windows (with WSL)
Python	Version 3.8 or higher
RAM	8 GB (32 GB recommended for large datasets)
Storage	20 GB free space
Processor	64-bit, 4 or more cores
GPU (optional)	NVIDIA GPU with 4+ GB VRAM (for PIDP acceleration)

A.2 Required Packages

```
numpy pandas scipy scikit-learn matplotlib seaborn psutil
biopython xgboost (optional but recommended)
metapredict aiupred (optional for PIDP)
```

A.3 Installation Steps

1. Clone the repository:

```
git clone https://github.com/polancodf2024/ebola-sgp-analysis.git
cd ebola-sgp-analysis
```

2. Install core dependencies:

```
pip install numpy pandas scipy scikit-learn matplotlib seaborn psutil
```

3. Install Biopython (for real instability index):

```
pip install biopython
```

4. Install XGBoost (optional, for improved ML):

```
pip install xgboost
```

5. (Optional) Install PIDP dependencies:

```
pip install metapredict aiupred
```

A.4 Configuration File Setup

The *config_EBOLA.json* file must be placed in the working directory. The system will automatically detect and load it. If *config_EBOLA.json* is not present, the system falls back to hardcoded default values.

Minimum structure of config.json for version 217.0:

```
{
  "metadata": {
    "version": "217.0",
    "schema_version": "3.4.0",
    "generated": "2026-09-12T00:00:00"
  },
  "operation_control": {
    "mode": "auto",
    "use_dummy_peptide": false,
    "dummy_sequence": "AAAAAAAAAAAA",
    "fallback_mode": "characterization",
    "evaluate_peptide": true,
    "use_bootstrap": true,
    "bootstrap_ci": 0.95,
    "apply_metric_weights": true,
    "apply_classification": true,
    "validate_target_ranges_in_characterization": true
  },
  "metric_weights": {
    "geodesic_distance": 0.25,
    "principal_angles": 0.10,
    "shannon_entropy": 0.15,
    "jensen_shannon": 0.15,
    "hellinger": 0.10,
    "gini": 0.10,
```

```

    "spearman": 0.10
  },
  "target_ranges": {
    "ebola": {
      "entropy": {"min": 0.5, "max": 1.0},
      "gini": {"min": 0.2, "max": 0.6},
      "grassmann_distance": {"min": 0.1, "max": 0.5}
    }
  },
  "classification_thresholds": {
    "excellent": 0.80,
    "good": 0.60,
    "moderate": 0.40,
    "poor": 0.00
  },
  "base_peptide_sequence": {
    "ebola": {
      "sequence": "GAAIGLAWIPYFGPAAEGI",
      "optimized_sequence": "GAAIGLAWIPYFGPAAEGT",
      "peptide_name": "EBOV-GP_inhibitor",
      "length": 19
    }
  },
  "processing_params": {
    "batch_size": 5000,
    "max_stored_proteins": 200,
    "n_bootstrap": 50,
    "max_workers": 4,
    "min_samples_per_group": 2
  },
  "data_paths": {
    "base_dir": "${SGPM_DATA}",
    "chembl_mapping": "${SGPM_DATA}/chembl_uniprot.txt",
    "dramp_tsv": "${SGPM_DATA}/general_amps.txt"
  }
}

```

A.5 Running *SGPMAIN* (version 217.0)

1. Place your FASTA files in the working directory:

```
zaire.unico.dat0  sudan.unico.dat0  reston.unico.dat0  ...
```

2. (Optional) Place `config_EBOLA.json` in the working directory.

3. Run the program:

```
python3 SGPMMAIN217.0v.py
```

4. Results are saved in:

```
results_{mode}_{YYYYMMDD_HHMMSS}/
```

A.6 Running the Head-to-Head Comparison

To reproduce the head-to-head comparison against BioPython ProtParam:

1. Ensure that BioPython is installed.
2. Ensure that the input FASTA file `todo_ebola.fasta` and the 30 reference files `*.unico.dat0` are in the ARCHIVOMAESTRO directory.
3. Run the comparison script:

```
python3 comparar_biopython_ebola.py
```

4. The script produces:
 - Per-virus directories `ebola_analysis_{virus}_{timestamp}/` with comparison CSVs, SGP metrics, and BioPython results.
 - Pairwise similarity matrices: `sgp_virus_matrix_*.csv` and `biopython_virus_matrix_*.csv`.
 - Combined comparison: `ebola_comparison_complete_*.csv`.
 - Final report: `FINAL_REPORT_*.txt`.

A.7 Quick Reference

```
# Run full analysis
python3 SGPMMAIN217.0v.py
```

```
# Run head-to-head comparison against BioPython ProtParam
python3 comparar_biopython_ebola.py
```

```
# View results
ls results_*/
```

```
# View summary
cat results_*/FINAL_SUMMARY_v2170.txt
```

```
# View metrics report
cat results_*/all_metrics_report_v2170.csv

# View base-vs-optimized comparison
cat results_*/base_vs_optimized_comparison.csv

# View ML CV metrics
cat results_*/ml_cv_metrics.csv
```

A.8 Operation Modes

Table A.2: Operation modes in **SGPMAIN** (version 217.0)

Mode	Description	When Used
Characterization	Only analyzes groups	No base peptide configured
Design	Only designs peptides	Base peptide configured, no analysis
Hybrid	Both analysis and design	Base peptide configured, evaluate peptide true
Auto	Automatically determined	Config specifies mode = "auto"

A.9 The 10 Core Metrics: Verification

Version 217.0 includes 17 unit tests for the 10 core metrics:

Table A.3: Unit tests for the 10 core metrics

Group	Test	Expected
Grassmann	Orthogonal angles	$\pi/2$
	Identical angles	0
	Geodesic identity	0
	Geodesic orthogonal	$(\pi/2)\sqrt{k}$
	Scalar curvature $Gr(1, 16)$	15
Information	Shannon deterministic	0
	Shannon uniform	2
	JS identical	0
	JS disjoint	1
	Hellinger identical	0
	Hellinger disjoint	1
Statistical	Gini uniform	0
	Gini [0,0,0,1]	0.5
	Spearman identical	1
Integration	Full comparison	Executes without error

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